

Mathematics IV - BSDS - 4th Sem. (Ordinary Differential Equations)

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Course Outline

- 1 First order differential equations, Picard's theorem, linear dependence and Wronskian.
- 2 Dimensionality of space of solutions, linear ODE with constant coefficients of second and higher order, Cauchy-Euler equations, Method of undetermined coefficients and method of variation of parameters.
- 3 Boundary Value Problems: System of linear differential equations with constant coefficients, fundamental matrix, matrix methods, Laplace Transform Method.

References

- ① Advanced Engineering Mathematics by Erwin Kreyszig
- ② Differential Equations by Shepley L. Ross

Introduction

- Differential equations constitute a large and very important branch of modern mathematics.
- Many physical laws and relations can be expressed mathematically in the form of differential equations.
- Our aim is the study of differential equations and the most important methods for solving them.

Differential Equations: Definition & Examples

Definition

An equation involving derivatives of one or more dependent variables with respect to one or more independent variables is called a **differential equation**.

In other words, A differential equation is an equation that involves differentials or differential coefficients.

Examples:

- ① $\frac{dy}{dx} = x + y$ (single independent & single dependent variables)
- ② $\frac{dy}{dx} + \frac{dz}{dx} = xyz$ (single independent & 2 dependent variables)
- ③ $y \frac{\partial u}{\partial x} + x \frac{\partial u}{\partial y} = 0$ (2 independent & single dependent variables)
- ④ $\frac{d^2y}{dx^2} + x \frac{dy}{dx} = 0$ (single independent & single dependent variables).

Ordinary Differential Equations

A Differential Equation involving ordinary derivatives of one or more dependent variables with respect to a single independent variable is called an **Ordinary Differential Equation**.

- In the example 1 the variable x is the single independent variable, and y is a dependent variable.
- In example 2 the independent variable is x , whereas y and z are both dependent.

Order of Differential Equations

The order of the highest ordered derivative involved in a differential equation is called the order of the differential equation.



$$\frac{d^2y}{dx^2} + xy \left(\frac{dy}{dx} \right)^2 = 0 : \text{Second Order ODE}$$



$$\frac{d^4x}{dt^4} + 5\frac{dx^2}{dt^2} + 3x = \sin t : \text{Fourth Order ODE}$$

Linear Ordinary Differential Equations

A linear ordinary differential equation of order n , in the dependent variable y and the independent variable x , is an equation that is in, or can be expressed in, the form

$$a_0(x) \frac{d^n y}{dx^n} + a_1(x) \frac{d^{n-1} y}{dx^{n-1}} + \cdots + a_{n-1}(x) \frac{dy}{dx} + a_n(x)y(x) = b(x)$$

where a_0 is not identically zero.

- Examples :

$$\frac{d^2 y}{dx^2} + 5 \frac{dy}{dx} + 6y = 0$$
$$\frac{d^4 y}{dx^4} + x^2 \frac{d^3 y}{dx^3} + x^3 \frac{dy}{dx} = xe^x$$

Non-linear Ordinary Differential Equations

A nonlinear ordinary differential equation is an ordinary differential equation that is not linear.

- Examples:

$$\frac{d^2y}{dx^2} + 5\left(\frac{dy}{dx}\right)^3 + 6y^2 = 0$$

$$\frac{d^2y}{dx^2} + 5y\left(\frac{dy}{dx}\right) + 6y = 0$$

Problems on Classification

- ① $\frac{d^3 y}{dx^3} + 4\frac{d^2 y}{dx^2} - 5\frac{dy}{dx} + 3y = \sin x$: Linear , 3rd Order ODE
- ② $\frac{d^4 y}{dx^4} + 3\left(\frac{d^2 y}{dx^2}\right)^5 + 5y = 0$: Non-linear , 4th Order ODE
- ③ $\frac{d^6 x}{dt^6} + \left(\frac{d^4 x}{dt^4}\right)\left(\frac{d^3 x}{dt^3}\right) + x = t$: Non-linear , 6th Order ODE
- ④ $\left(\frac{dr}{ds}\right)^3 = \sqrt{\frac{d^2 r}{ds^2} + 1}$
 $\implies \frac{d^2 r}{ds^2} - \left(\frac{dr}{ds}\right)^6 + 1 = 0$: Linear , 2nd Order ODE

First Order ODE

- First-order ODEs contain only the first derivative $y' \equiv \frac{dy}{dx}$ and may contain y and any given functions of x . Hence we can write them as

$$F(x, y, y') = 0 : \text{Implicit Form} \quad (0.1)$$

- often re-written in the form

$$y' = f(y(x), x) : \text{Explicit Form}$$

Solution of a Differential Equation

A function $y = h(x)$ is called a solution of a given ODE $F(x, y, y') = 0$ on some open interval if it is defined and differentiable throughout the interval and is such that the equation becomes an identity if y and y' are replaced with h and h' , respectively.

For an n th-order ordinary differential equation $F = \left[x, y, \dots, \frac{d^n x}{dy^n} \right] = 0$, Let f be a real function defined for all x in a real interval I then it is called a solution of the differential equation on I if

- ① $F[x, f(x), f'(x), \dots, f^{(n)}(x)]$ is defined for all $x \in I$
- ② $F[x, f(x), f'(x), \dots, f^{(n)}(x)] = 0$

Verification of Solutions

- Consider the ODE : $x \frac{dy}{dx} + y = 0$
 - Verify that $y = c/x$ is a solution of the ODE
 - Differentiating $y = c/x$ we get, $\frac{dy}{dx} = -\frac{c}{x^2}$
 - Multiply this by x , one obtains $x \frac{dy}{dx} = -\frac{c}{x} = -y$; thus, the given ODE is formulated.
- The function $f(x) = 2 \sin(x) + 3 \cos(x)$ is a solution of the differential equation $\frac{d^2y}{dx^2} + y = 0$ (verify it yourselves !)
- The relation $x^2 + y^2 = 25$ is a solution for the differential equation $x + y \frac{dy}{dx} = 0$ (Check !)

Solution by Calculus

Solution Curves

- Consider the first-order differential equation $\frac{dy}{dx} = 2x$
- The function f_0 defined for all real x by $f_0(x) = x^2$ is a solution of this equation.
- So also are the functions f_1 , f_2 , and f_3 defined for all real x by $f_1(x) = x^2 + 1$, $f_2(x) = x^2 + 2$ and $f_3(x) = x^2 + 3$
- for each real number c , the function f_c defined for all real x by

$$f_c(x) = x^2 + c$$

is a solution of the differential equation.

- In other words, this solution $f_c(x)$ defines an infinite family of functions, one for each real constant c , and every function of this family is a solution of the ODE.

General Solution

- We see that the solution of each ODE contains an arbitrary constant c .
- Such a solution containing an arbitrary constant c is called a **General Solution**.
- Geometrically, the general solution of an ODE is a family of infinitely many solution curves, one for each value of the constant c .

Particular Solution

- If we choose a specific c we obtain what is called a **Particular Solution**.
- A particular solution does not contain any arbitrary constants.
- In most cases, general solutions exist, and every solution not containing an arbitrary constant is obtained as a particular solution by assigning a suitable value to c .

Initial Value Problem

- In most cases the unique solution of a given problem, hence a particular solution, is obtained from a general solution by an initial condition with given values.
- Those values are used to determine a value of the arbitrary constant c .
- Geometrically this condition means that the solution curve should pass through the point (initial values as co-ordinates) in the xy -plane.
- An ODE, together with an initial condition, is called an initial value problem.
- The **Initial Value Problem (IVP)** is of the form

$$\frac{dy}{dx} = f(x, y) \quad , \quad y_0 = f(x_0)$$

Examples of IVP

- The IVP

$$\frac{dy}{dx} = 3y \quad ; \quad y(0) = 5.7$$

- General solution: $y(x) = ce^{3x}$
- Given initial condition: $y(0) = 5.7$
- Compute constant: $c = 5.7$
- Particular solution: $y(x) = 5.7e^{3x}$

Problem 10: Verification and Particular Solution

Given:

$$y' + 5xy = 0, \quad y = ce^{-2.5x^2}, \quad y(0) = \pi$$

(a) Verify that y is a solution:

- Start with $y = ce^{-2.5x^2}$
- Differentiate:

$$y' = ce^{-2.5x^2}(-5x)$$

- Factor:

$$y' = -5xce^{-2.5x^2}$$

- Compute $5xy$:

$$5xy = 5xce^{-2.5x^2}$$

- Substitute into the ODE:

$$y' + 5xy = -5xce^{-2.5x^2} + 5xce^{-2.5x^2} = 0$$

- Hence, $y = ce^{-2.5x^2}$ satisfies the differential equation.

(b) Find the particular solution using $y(0) = \pi$:

- Substitute $x = 0$ into $y = ce^{-2.5x^2}$:

$$y(0) = ce^0 = c$$

- Given $y(0) = \pi$, we get:

$$c = \pi$$

- Therefore, the particular solution is:

$$y(x) = \pi e^{-2.5x^2}$$

Problem 14: Verification and Particular Solution

Given:

$$y' \tan x = 2y - 8, \quad y = c \sin^2 x + 4, \quad y\left(\frac{\pi}{2}\right) = 0$$

(a) Verify that y is a solution:

- Start with the given function:

$$y = c \sin^2 x + 4$$

- Differentiate:

$$y' = c \cdot 2 \sin x \cos x = 2c \sin x \cos x$$

- Multiply by $\tan x$:

$$y' \tan x = 2c \sin x \cos x \cdot \frac{\sin x}{\cos x} = 2c \sin^2 x$$

- Compute the right-hand side:

$$2y - 8 = 2(c \sin^2 x + 4) - 8 = 2c \sin^2 x$$

- Since both sides are equal, $y = c \sin^2 x + 4$ satisfies the ODE.

(b) Find the particular solution using $y(\pi/2) = 0$:

- Substitute $x = \pi/2$ into $y = c \sin^2 x + 4$:

$$0 = c \sin^2\left(\frac{\pi}{2}\right) + 4$$

- Since $\sin(\pi/2) = 1$, this gives:

$$0 = c + 4 \Rightarrow c = -4$$

- Therefore, the particular solution is:

$$y(x) = 4 - 4 \sin^2 x$$

Exact Differential Equations

- A first-order differential equation can be written in:
 - **Derivative form:** $\frac{dy}{dx} = f(x, y)$
 - **Differential form:** $M(x, y) dx + N(x, y) dy = 0$
- The expression $M(x, y) dx + N(x, y) dy$ is called an **exact differential** in a domain D if there exists a function $F(x, y)$ such that:

$$dF = M(x, y) dx + N(x, y) dy$$

- This means:

$$\frac{\partial F(x, y)}{\partial x} = M(x, y), \quad \frac{\partial F(x, y)}{\partial y} = N(x, y)$$

- If $M(x, y) dx + N(x, y) dy$ is exact, then the equation

$$M(x, y) dx + N(x, y) dy = 0$$

is called an **Exact Differential Equation**.

Theorem: Test for Exact Differential Equations

- Consider the differential equation:

$$M(x, y) dx + N(x, y) dy = 0,$$

where M and N have continuous first partial derivatives in a rectangular domain D .

- If the equation is exact in D , then:**

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x} \quad \text{for all } (x, y) \in D.$$

- Conversely, if**

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x} \quad \text{for all } (x, y) \in D,$$

then the equation is exact in D .

- Remark:** Proof of this theorem is beyond the scope of the current class.

Examples: Testing Exact Differential Equations

Example 1:

$$y^2 dx + 2xy dy = 0$$

- Identify:

$$M(x, y) = y^2, \quad N(x, y) = 2xy$$

- Compute partial derivatives:

$$\frac{\partial M}{\partial y} = 2y, \quad \frac{\partial N}{\partial x} = 2y$$

- Since $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$ for all (x, y) , the equation is **exact**.
- Hence, it is exact in every rectangular domain D .

Example 2:

$$y \, dx + 2x \, dy = 0$$

- Identify:

$$M(x, y) = y, \quad N(x, y) = 2x$$

- Compute partial derivatives:

$$\frac{\partial M}{\partial y} = 1, \quad \frac{\partial N}{\partial x} = 2$$

- Since $\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$, the equation is **not exact**.
- Hence, it is not exact in any rectangular domain D .

Solution of Exact Differential Equations

- Consider the differential equation

$$M(x, y) dx + N(x, y) dy = 0$$

which is exact in a rectangular domain D .

- Exactness implies the existence of a function $F(x, y)$ such that

$$\frac{\partial F}{\partial x} = M(x, y), \quad \frac{\partial F}{\partial y} = N(x, y)$$

- Using the total differential,

$$dF(x, y) = \frac{\partial F}{\partial x} dx + \frac{\partial F}{\partial y} dy$$

the given equation can be written as

$$dF(x, y) = 0$$

- Integrating, we obtain the general solution

$$F(x, y) = c$$

where c is an arbitrary constant.

Example: Solving an Exact Differential Equation

- Solve the differential equation:

$$(3x^2 + 4xy) dx + (2x^2 + 2y) dy = 0$$

- Identify:

$$M(x, y) = 3x^2 + 4xy, \quad N(x, y) = 2x^2 + 2y$$

- Check exactness:

$$\frac{\partial M}{\partial y} = 4x, \quad \frac{\partial N}{\partial x} = 4x$$

Since these are equal, the equation is **exact**.

- Find the potential function $F(x, y)$:

$$\frac{\partial F(x, y)}{\partial x} = M(x, y) \Rightarrow F(x, y) = \int (3x^2 + 4xy) dx$$

$$F(x, y) = x^3 + 2x^2y + \phi(y)$$

- Differentiate with respect to y :

$$\frac{\partial F(x, y)}{\partial y} = 2x^2 + \frac{d\phi}{dy}$$

Set $\frac{\partial F(x, y)}{\partial y} = N(x, y) = 2x^2 + 2y$.

- Hence,

$$\frac{d\phi}{dy} = 2y \Rightarrow \phi(y) = y^2 + c$$

- Final solution:

$$F(x, y) = x^3 + 2x^2y + y^2 + c_0$$

- Hence a one-parameter family of solution is $F(x, y) = C_1$, or

$$x^3 + 2x^2y + y^2 + c_0 = c_1$$

. Combining the constants c_0 and c_1 we may write this solution as

$$x^3 + 2x^2y + y^2 = c$$

Solving an Exact Differential Equation (IVP)

Given initial-value problem:

$$(2x \cos y + 3x^2 y) dx + (x^3 - x^2 \sin y - y) dy = 0, \quad y(0) = 2$$

- Identify:

$$M(x, y) = 2x \cos y + 3x^2 y, \quad N(x, y) = x^3 - x^2 \sin y - y$$

- Compute partial derivatives:

$$\frac{\partial M}{\partial y} = -2x \sin y + 3x^2, \quad \frac{\partial N}{\partial x} = 3x^2 - 2x \sin y$$

- Since $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$, the equation is **exact** in every rectangular domain.

Step 1: Find the potential function $F(x, y)$

- We seek $F(x, y)$ such that:

$$\frac{\partial F}{\partial x} = M(x, y), \quad \frac{\partial F}{\partial y} = N(x, y)$$

- Integrate $M(x, y)$ with respect to x :

$$F(x, y) = \int (2x \cos y + 3x^2 y) dx + \phi(y)$$

- This gives:

$$F(x, y) = x^2 \cos y + x^3 y + \phi(y)$$

Step 2: Determine $\phi(y)$

- Differentiate $F(x, y)$ with respect to y :

$$\frac{\partial F}{\partial y} = -x^2 \sin y + x^3 + \phi'(y)$$

- Set this equal to $N(x, y) = x^3 - x^2 \sin y - y$:

$$-x^2 \sin y + x^3 + \phi'(y) = x^3 - x^2 \sin y - y$$

- Cancel common terms to get:

$$\phi'(y) = -y$$

- Integrate:

$$\phi(y) = -\frac{y^2}{2} + C$$

Step 3: Write the implicit solution

- Substitute into $F(x, y)$:

$$F(x, y) = x^2 \cos y + x^3 y - \frac{y^2}{2} + C$$

- Hence, the general solution is:

$$x^2 \cos y + x^3 y - \frac{y^2}{2} = C$$

Step 4: Apply the initial condition $y(0) = 2$

- Substitute $x = 0, y = 2$:

$$C = 0 + 0 - \frac{4}{2} = -2$$

- Therefore, the solution of the IVP is:

$$x^2 \cos y + x^3 y - \frac{y^2}{2} = -2$$